

Course Activities - Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Aligned deck: Trainer Slides v7.0

Each activity has its own self-contained folder with a procedure, scenario, worksheet, evidence checklist and same-folder printable PDFs.

1. [Device Physics Evidence Map](#) - K1 | A1 - 35 minutes - concepts 18-105 - overview slide 441
2. [Process Module Architecture](#) - A1 - 40 minutes - concepts 120-258 - overview slide 442
3. [CMOS Manufacturing Flow and Control Gates](#) - A2 - 50 minutes - concepts 259-298 - overview slide 443
4. [Lithography-Etch Interaction Case](#) - A3 - 45 minutes - concepts 141-217 - overview slide 444
5. [Doping and Thermal-Budget Interaction](#) - A3 - 45 minutes - concepts 218-237 - overview slide 445
6. [Yield and Performance Metric Verification](#) - K2 | A4 - 55 minutes - concepts 316-373 - overview slide 446
7. [Reliability Failure Analysis](#) - K2 | A4 - 45 minutes - concepts 374-385 - overview slide 447
8. [Follow-Up Action and Integration Review](#) - A5 - 60 minutes - concepts 420-440 - overview slide 448

Assessment alignment

- Activities 2-3 prepare evidence for Case Study Task 1 (A1-A2).
- Activities 4-8 prepare evidence for Case Study Task 2 (A3-A5).
- Activity 1 reinforces K1 device-physics reasoning used in the Written Assessment.
- Activities 6-7 reinforce K2 performance metrics used in the Written Assessment.